

YUVRAJ GREWAL

Electrical Engineering Technology Graduate

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PROFESSIONAL SUMMARY

Electrical Engineering Technology graduate from the Northern Alberta Institute of Technology (Dean's Honor List) with hands-on experience across protective relay testing and commissioning, Allen-Bradley PLC and VFD programming, SCADA and HMI development, industrial networking, and electrical design with AutoCAD, Revit, and ETAP. Comfortable bridging design and field work — from full IFC drawing packages and load-flow analysis to live equipment commissioning and real-time monitoring systems. Eligible for ASET certification (C.E.T.).

TECHNICAL SKILLS

Relay & Protection	MANTA MTS-5100 test set; SEL AcSElerator; electromechanical & numerical relays (Westinghouse CO, GE-735/750/469, Basler BE1-951, SEL-751, MiCOM P123); protection functions 50, 51, 67, 87; metering, harmonic blocking; GOOSE messaging (IEC 61850)
Apparatus Testing	Insulation resistance, DC Hi-Pot, TTR, winding resistance, polarization index, capacitance/dissipation factor, breaker timing, dielectric oil testing — MV transformers, CTs/PTs, cables, breakers
PLC & SCADA	Allen-Bradley ControlLogix 5580 / CompactLogix; Studio 5000; PowerFlex 70 & 753 VFD commissioning; Ignition SCADA with Perspective HMI
Networking	TCP/IP, EtherNet/IP, Modbus; VLAN & IP addressing; firewall policies, static routing, 1:1 NAT; Phoenix Contact routers; Cisco Packet Tracer
Design & CAD	AutoCAD — SLDs, AC/DC three-line diagrams, DC schematics, wiring diagrams, panel layouts; Revit 3D modelling; ETAP load flow & fault analysis
Standards	NETA acceptance testing; CEC, CSA, IEC, IEEE; NERC/CIP awareness; substation safety (EPCOR certified)
Software	Microsoft Word, Excel, Outlook; technical test sheets & commissioning report preparation
Languages	English, Hindi, Punjabi

TECHNICAL PROJECTS

Industrial Ethernet Network Design & SCADA Automation · Ranked #1

2025

Northern Alberta Institute of Technology — Final Project · Team of 2

- Ranked #1 among all final projects. Designed and implemented a segmented industrial network (Plant, SCADA, WAN zones) using Phoenix Contact MGuard 2102 and TC Router 3002T-4G with firewall policies, static routing, and 1:1 NAT.
- Integrated nine PowerFlex 70/753 VFDs with dual ControlLogix 5580 PLCs via EtherNet/IP, including PLC-to-PLC communication as a workaround for a Studio 5000 limitation above five VFDs.
- Developed an Ignition SCADA system with a Perspective HMI dashboard for real-time monitoring of voltage, current, frequency, and power.

IFC Electrical Design — Industrial Plant Package

2025

Northern Alberta Institute of Technology — IFC Design Project

- Produced a full Issued-for-Construction electrical package (13 drawings): MCC single-line diagram (8 motors, 25–75 HP, with VFD/ESS/FVNR starters), hazardous area classification (Class 1 Zone 2), cable tray layout, panel schedules, and MCC schedule.
- Completed CEC Rules 28-106 / 28-306 motor branch circuit calculations, ETAP load flow analysis, and 3D Revit cable tray modelling.

Protective Relay Testing & Commissioning

2024 – 2025

Northern Alberta Institute of Technology — Apparatus & Relay Testing Lab

- Tested and commissioned electromechanical and digital protective relays (Westinghouse CO, GE-735/750/469, Basler BE1-951, SEL-751) using the MANTA MTS-5100 per NETA acceptance testing standards.
- Executed protection element testing for functions 50, 51, 67, 87, metering, and harmonic blocking; prepared detailed test sheets and commissioning reports to manufacturer specifications.
- Configured and validated GOOSE messaging between SEL relays using SEL AcSELeator, including VLAN and IP addressing for IEC 61850 communication.

Protective Relay Replacement — Brownfield Design

2025

Northern Alberta Institute of Technology — Capstone Project

- Supported an upgrade from MiCOM P123 to SEL-751; reviewed relay I/O specifications and revised full AutoCAD drawing packages including SLDs, AC/DC three-line diagrams, DC schematics, wiring diagrams, and relay panel layouts.
- Traced existing protection schematics and analysed as-built documentation to ensure accurate integration; delivered a revised drawing package supporting installation and commissioning.

Electrical Apparatus Testing & Preventive Maintenance

2024 – 2025

Northern Alberta Institute of Technology — Apparatus Lab

- Inspected, tested, and maintained medium-voltage equipment (transformers, CTs/PTs, cables, circuit breakers) per NETA using insulation resistance, DC Hi-Pot, TTR, winding resistance, polarization index, dissipation factor, and breaker timing tests.
- Conducted dielectric oil testing and equipment condition assessments; produced detailed field test reports evaluated against NETA standards and manufacturer specifications.

WORK EXPERIENCE

Sales Associate

Dec 2024 – Oct 2025

ACE Liquor Discounter — Edmonton, AB

- Delivered professional customer service and resolved issues effectively; built strong communication and interpersonal skills in a fast-paced, high-volume environment.
- Handled accurate POS transactions, maintained organized inventory, and upheld store safety and security procedures.

EDUCATION

Electrical Engineering Technology Diploma

Jan 2024 – Dec 2025

Northern Alberta Institute of Technology — Edmonton, AB · Dean's Honor List

- Eligible for ASET certification (C.E.T.). Coursework: Apparatus & Relay Testing (NETA), Allen-Bradley PLC & VFD Programming, AutoCAD & Revit Design, Power Systems Analysis (ETAP), Industrial Networking & SCADA.

CERTIFICATIONS & LICENCES

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| • EPCOR Electricity/Technologies Safety Orientation (Substations) | • Class 5 Driver's Licence |
| • Standard First Aid & CPR | • ProServe |
| • WHMIS | • Fire Safety & Extinguishers |

EXTRACURRICULARS

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| • VP of Finance — EET Program Club (NAIT) | • Captain — High School Basketball Team |
| • Open House Representative (NAIT) | |